P53657 · KPYR_MOUSE

Protein[i]
Pyruvate kinase PKLR

Gene[i]
Pklr

Status[i]
UniProtKB reviewed (Swiss-Prot)

Organism[i]
Mus musculus (Mouse)

Amino acids
574 (go to sequence)

Protein existence[i]
Evidence at protein level

Annotation score[i]
5/5

Function[i]

Pyruvate kinase that catalyzes the conversion of phosphoenolpyruvate to pyruvate with the synthesis of ATP, and which plays a key role in glycolysis. Also produces the side product 2-phospholactate which can inhibit fructose-2,6-bisphosphate production. 2-phospholactate can be dephosphorylated by PGP which prevents the inhibition of fructose-2,6-bisphosphate production and allows glycolysis to occur. By Similarity

Miscellaneous

There are 4 isozymes of pyruvate kinase in mammals: L, R, M1 and M2. L type is major isozyme in the liver, R is found in red cells, M1 is the main form in muscle, heart and brain, and M2 is found in early fetal tissues.
Curated

Catalytic activity[i]

Rhea:18157

pyruvate + ATP = phosphoenolpyruvate + ADP + H^+ By Similarity
This reaction proceeds in the backward direction. By Similarity
EC:2.7.1.40 (UniProtKB | ENZYME | Rhea)

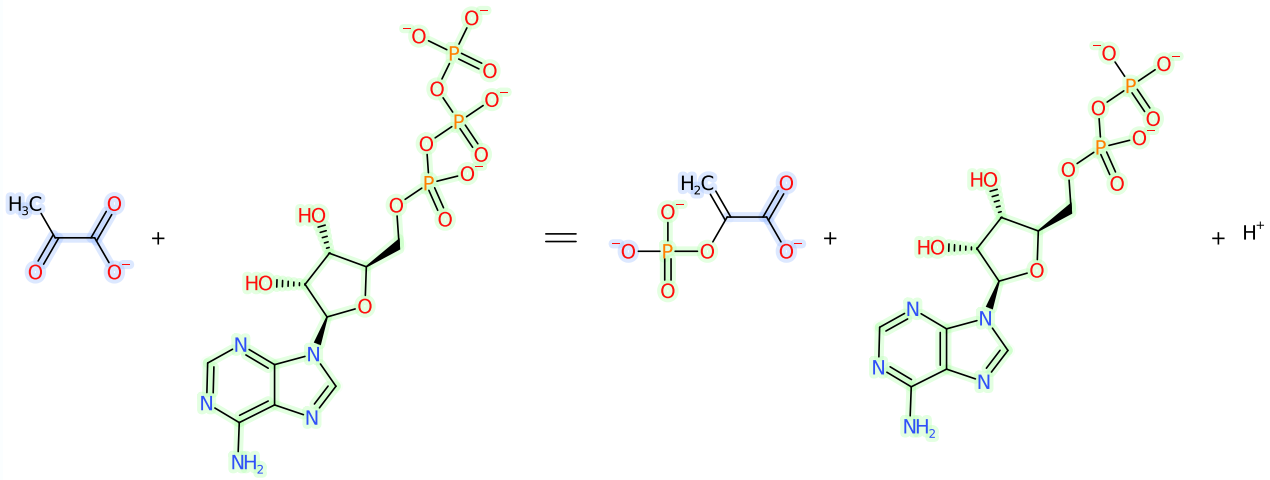
Hide reaction details

Reaction | Atom map

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I agree, dismiss this banner

Color atoms ☑, Number atoms ☐, Confidence = 0.34795



pyruvate » ATP » phosphoenolpyruvate » ADP » H^+ »

Rhea:86567

(S)-lactate + ATP = (2S)-2-phospholactate + ADP + H^+ By Similarity

View reaction details

Cofactor[i]

Protein has several cofactor binding sites:

Mg^{2+} (UniProtKB | Rhea | CHEBI:18420) By Similarity
Mn^{2+} (UniProtKB | Rhea | CHEBI:29035) By Similarity

K^+ (UniProtKB | Rhea | CHEBI:29103) By Similarity

Activity regulation[i]

Allosterically activated by fructose 1,6-bisphosphate. By Similarity

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Features

Showing features for binding site[i], site[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Binding site		116	substrate By Similarity	
+	Binding site		118	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		118-121	ATP (UniProtKB | ChEBI) By Similarity	Tools Add
+	Binding site		120	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		156	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		157	K^+ (UniProtKB | ChEBI) By Similarity	

Expand table

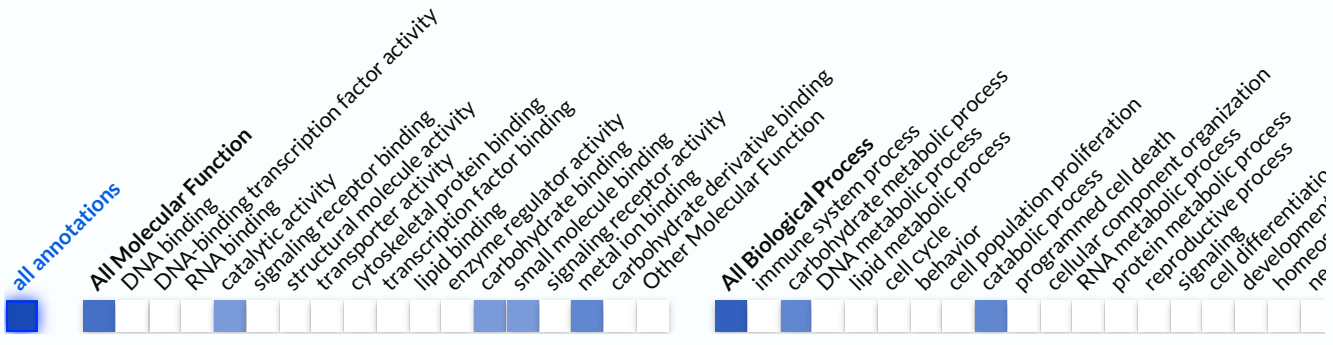
Gene Ontology[i]

GO annotations | GO-CAM models New

Gene Ontology (GO) annotations organized by slimming set.

Slimming set:

Animals and fungi (Opisthokonta)



Cell color indicative of number of GO terms

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ASPECT	TERM
Molecular Function	monosaccharide binding Source:GO_Central
Molecular Function	potassium ion binding Source:InterPro
Molecular Function	pyruvate kinase activity Source:MGI 2 Publications
Biological Process	cellular response to insulin stimulus Source:GO_Central
Biological Process	glycolytic process Source:MGI 1 Publication
Biological Process	pyruvate biosynthetic process Source:GO_Central
Biological Process	response to ATP Source:GO_Central
Biological Process	response to cAMP Source:GO_Central
Biological Process	response to other organism Source:MGI 1 Publication

Access the complete set of GO annotations on QuickGO

Keywords[i]

Molecular function

#Allosteric enzyme
#Kinase
#Transferase

Biological process

#Glycolysis

Ligand

#ATP-binding
#Magnesium
#Manganese
#Metal-binding
#Nucleotide-binding

More keywords

Enzyme and pathway databases

Reactome

R-MMU-70171 Glycolysis
R-MMU-70268 Pyruvate metabolism

SABIO-RK

P53657

UniPathway

UPA00109UER00188

ENZYME

Search...

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Recommended name
Pyruvate kinase PKLR

EC number
EC:2.7.1.40 (UniProtKB | ENZYME | Rhea) By Similarity

Alternative names
L-PK
Pyruvate kinase isozymes L/R

Gene names[i]

Name
Pklr

Organism names

Taxonomic identifier[i]
10090 (NCBI)

Organism[i]
Mus musculus (Mouse)

Strain
CBA/N

Taxonomic lineage[i]
Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Euarchontoglires > Glires > Rodentia > Myomorpha > Muroidea > Muridae > Murinae > Mus > Mus

Accessions[i]

Primary accession
P53657

Proteomes[i]

Identifier
UP000000589

Component[i]
Unplaced

Organism-specific databases

AGR
MGI:97604

MGI
MGI:97604 Pklr

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Subcellular Location[i]

UniProt Annotation | GO Annotation

cytoplasm Source:MGI 1 Publication
cytosol Source:MGI 1 Publication
plasma membrane Source:MGI 1 Publication
pyruvate kinase complex Source:GO_Central

Complete GO annotation on QuickGO

Interaction[i]

Subunit[i]

Homotetramer. By Similarity

Complex viewer[i]

Select complex

CPX-3097 PKL pyruvate kinase complex

View interactors in UniProtKB
View CPX-3097 in Complex Portal

Protein-protein interaction databases

ComplexPortal
CPX-3097 PKL pyruvate kinase complex [P53657-2]
CPX-3098 PKR pyruvate kinase complex [P53657-1]

FunCoup
P53657 309 interactors

STRING
10090.ENSMUSP00000035417

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RNAct
P53657 protein

Structure[i]



3D structure databases

AlphaFoldDB
P53657

ModBase
Search...

SMR
P53657

Family & Domains[i]

Sequence similarities[i]

Belongs to the pyruvate kinase family. Curated

Family and domain databases

- View the Phylogenomic databases for this entry within the Homologs section.
- View all family and domain features for this entry's canonical sequence in the UniParc Feature Viewer.

CDD
cd00288 Pyruvate_Kinase 1 hit

FunFam
2.40.33.10:FF:000001 Pyruvate kinase 1 hit
3.20.20.60:FF:000025 Pyruvate kinase 1 hit
3.40.1380.20:FF:000001 Pyruvate kinase 1 hit

Gene3D
3.20.20.60 Phosphoenolpyruvate-binding domains 1 hit
2.40.33.10 PK beta-barrel domain-like 1 hit
3.40.1380.20 Pyruvate kinase, C-terminal domain 1

InterPro
View protein in InterPro
IPR001697 Pyr_Knase
IPR015813 Pyrv/PenolPyrv_kinase-like_dom
IPR040442 Pyrv_kinase-like_dom_sf
IPR011037 Pyrv_Knase-like_insert_dom_sf
More InterPro links

NCBIfam
NF004491 PRK05826.1 1 hit
NF004978 PRK06354.1 1 hit
TIGR01064 pyruv_kin 1 hit

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PRINTS
PR01050 PYRUVTKNASE

PROSITE
View protein in PROSITE
PS00110 PYRUVATE_KINASE 1 hit

Pfam
View protein in Pfam
PF00224 PK 1 hit
PF02887 PK_C 1 hit

SUPFAM
SSF51621 Phosphoenolpyruvate/pyruvate domain 1 hit
SSF50800 PK beta-barrel domain-like 1 hit
SSF52935 PK C-terminal domain-like 1 hit

MobiDB
Search...

Sequence & Isoform[i]

Align isoforms (2) Add isoforms

Sequence status[i]
Complete

This entry describes **2** isoforms[i] produced by **Alternative splicing**.

P53657-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name
R-type
Synonyms
PKR

See also
sequence in **UniParc** or sequence clusters in **UniRef**

Length
574

Mass (Da)
62,309

Last updated
1996-10-01 v1

MD5 Checksum[i]
F2ABFCE70A9D37BD1DBAE160CE6DC209

MSVQENELPQQLWPWIFKSQKDLAKSALSGAPGGPAGYLRRASVAQLTQELGTAFFQQQQLPAAMADTFLEHLC
LLDIDSEPVAARSTSIIATIGPASRSVDRLKEMIKAGMNIARLNFSHGSHEYHAESIANIREAAESFATSPLSYRPVAIAL

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ETSDVANAVLDGADCIMLSGETAKGSFPVEAVKMQHAIAREAEAAVYHRQLFEELRRAAPLSRDPTEVTAIGAVEA
SFKCCAAAIIVLTKTGRSAQLLSRYRPRAAVIAVTRSAQAARQVHLSRGVFPLLYREPPEAVWADDVDRRVQFGIES
GKLRGFLRVGDLVIVVTGWRPGSGYTNIMRVLTIS

P53657-2

Name	Synonyms
L-type	PKL

Computationally mapped potential isoform sequences[i]

There are 3 potential isoforms mapped to this entry

Align Add View all

Entry	Entry name	Gene name	Length
E9Q509	E9Q509_MOUSE	Pklr	543
D3Z2C4	D3Z2C4_MOUSE	Pklr	72
G3X925	G3X925_MOUSE	Pklr	574

Keywords[i]

Coding sequence diversity

#Alternative splicing

Technical term

#Reference proteome

Sequence databases

CCDS

CCDS17489.1 [P53657-1]

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
S79731 EMBL · GenBank · DDBJ	AAB35435.1 EMBL · GenBank · DDBJ	mRNA	
D63764 EMBL · GenBank · DDBJ	BAA23642.1 EMBL · GenBank · DDBJ	mRNA	

Similar Proteins[i]

UniRef clusters[i]

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Homologs[i]

Orthologs & paralogs[i]



Phylogenomic databases

InParanoid
P53657

PhylomeDB
P53657

eggNOG
KOG2323 Eukaryota

Organism-specific databases

AGR
MGI:97604

Disclaimer

Any medical or genetic information present in this entry is provided for research, educational and informational purposes only. It is not in any way intended to be used as a substitute for professional medical advice, diagnosis, treatment or care. Our staff consists of biologists and biochemists that are not trained to give medical advice.

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